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DATAGENT: Data Analytics Toolkit for Automatic Generation of Dashboards and Tailored Insights

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Project Guide
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Abstract

- Businesses often face challenges in converting raw datasets and requirements into actionable insights due to time-consuming data cleaning, feature engineering, and analysis processes. This project presents an AI-powered Data Analyst Agent that automates the entire workflow from understanding business requirements and processing datasets to performing feature selection and generating analysis outputs in Python, R, SQL, Excel or SQL + Python. The agent mimics a human data analyst, creating dashboards, queries, and reports efficiently, while allowing iterative refinements and providing downloadable outputs.

Sustainable Development Goals(SDG) mapped

- **SDG 4 (Quality Education):** Teaches data analysis through auto-generated Python, SQL, and Excel workflows. Helps learners understand business problem-solving via editable, real-world dashboards.

Introduction

- In today's data-driven world, organizations often struggle to extract meaningful insights from raw datasets and business requirements.
- For a seamless and automated data analysis experience, our AI Data Analyst Agent integrates:

➢ Understanding business requirements and recommended analysis questions.

➢ Dataset cleaning and preprocessing for small to large datasets.

➢ Feature engineering and selection based on the analysis requirements.

➢ Analysis and dashboard creation in Python, R, SQL, Excel, or SQL+Python.

➢ Iterative refinements and downloadable code, queries, and dashboards

Motivation

- Businesses spend significant time and resources on manual data cleaning, feature selection, and report generation.
- Existing tools either handle only parts of the workflow or require advanced technical knowledge.
- There is a need for an AI-powered agent that automates the entire data analysis workflow like a real data analyst.
- Our goal is to make data analysis faster, accurate, and accessible for organizations of all sizes.

Objectives

- To architect a high-performance data cleaning pipeline using Polars and Pandas for efficiency, integrating Random Forest and KNN to drive an adaptive, context-aware imputation strategy that outperforms standard rule-based methods.
- To engineer a semantic feature selection system that combines the statistical rigor of NumPy and Polars with the reasoning capabilities of Large Language Models (Gemini) to automatically filter features based on business utility rather than just variance.
- To automate the generation of analysis-ready scripts in Python, R, SQL using LLM and Excel pivot logic using Pandas and XlsxWriter, tailored to business requirements.
- To create editable and downloadable Excel dashboards using XlsxWriter, OpenpyExcel and Pywin32, including pivot tables, charts, and suggestions for easy customization.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Automated Data Processing and Feature Engineering for Deep Learning and Big Data Applications: A Survey	Alhassan Mumuni, Fuseini Mumuni	2025	Survey of automated preprocessing, augmentation, feature engineering, AutoML methods (GANs, VAEs, Auto-Sklearn).	Limited automation, scalability issues, poor generalization, lack of explainability, task support limitations.
2	Natural Language Interface for Data Visualization	Buchepalli Praneeth, Mohana, Ashutosh Kumar Singh, Suma B, Jyothika K Raju	2024	Visualization tool using LLMs (Mistral 7B, LLaMa3 8B, GPT-3.5-turbo) + LIDA for data loading, summarization, goal generation, visualization.	Ambiguity in queries, dependence on LLMs, no real-time data, limited customization, weak robustness, and privacy concerns.
3	DeepEye: Towards Automatic Data Visualization	Yuyu Luo, Xuedi Qin, Nan Tang, Guoliang Li	2018	ML models (Decision Tree, LambdaMART) + expert rules for visualization selection.	Dependent on training data, limited chart types, scalability issues, no NL interaction.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	AI Powered SQL Query Generator and Data Visualizer	Girish R., Swasthik K., Veera Mani Kandan S., Muralidharan S., Vinoth M., Surendar Prakash T.	2024	NLP-based SQL query generation, AI-optimized queries, MySQL/PostgreSQL integration, auto dashboards.	Conceptual MVP, lacks scalability, basic visualizations, no enterprise integration, monolingual NLP.
5	Chat2VIS: Generating Data Visualizations via NL Using ChatGPT, Codex, GPT-3	Paula Maddigan, Teo Susnjak	2023	Chat2VIS system using GPT-3, Codex, ChatGPT + prompt engineering to generate visualizations.	Aesthetic control issues, variability in outputs, limited evaluation, lacks benchmarks.
6	AI-Powered Visual Analytics: Effortless Data Insights	Devi Dharshini S S, Geethanjali M, Saran M	2025	Together AI LLMs + Pandas, Plotly, Seaborn, Matplotlib + Streamlit for NL queries to real-time visualization.	Ambiguous query handling, limited customization, weak error handling, lacks voice input & dashboarding.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Analysis of NLP for Code Generation using COPRAS	Madhusudhan Dasari Sreeramulu	2024	COPRAS applied to Seq2Seq, SNM, Tree2Tree, TRANX, Coarse-to-Fine ranking.	No data transparency, oversimplified metrics, lacks domain-specific context.
8	Leveraging LLMs for Data Analysis Automation	Jacqueline A. Jansen, Artür Manukyan, Nour Al Khoury, Altuna Akalin	2025	R package mergen + GPT-3.5/GPT-4, prompt engineering, self-correction, applied to bioinformatics tasks.	Executable code not always correct, poor on complex tasks, privacy issues.
9	Plotplay: Automated Data Visualization Website using Python and Plotly	Zarinabegam K. Mundargi, Rohan More, Krisha Patel, Sudhanshu Pathrabe, Aayushi Patel, Shreyash Patil	2023	Flask-based tool with Pandas + Plotly (10 chart types).	CSV-only, no preprocessing, limited plots, no customization.

Research Gap(Limitations of existing systems)

- **Fragmented Solutions:** Most tools address only one part of the workflow, lacking a complete end-to-end automation from raw data to dashboard.
- **Shallow Automation:** Existing systems cannot intelligently clean data, engineer features, or select relevant columns based on analytical goals.
- **Limited User Interaction:** Users cannot iteratively provide feedback or customize outputs after initial analysis.
- **Platform Restriction:** Tools are often tied to a single output format (Python, R, SQL, or Excel), limiting flexibility for different user needs.
- **Poor Handling of Real-World Data:** Many systems fail to manage large, complex, or messy datasets effectively, making them unsuitable for enterprise-scale applications.

Problem Definition

- In today's data-driven world, organizations face major challenges in transforming raw data into actionable insights. Messy and inconsistent datasets, irrelevant features, the complexity of using multiple platforms (Python, R, SQL, Excel), and manual dashboard creation slow down decision-making and limit accessibility for non-technical users.
- Problem Statement: There is a need for an AI-powered system that can automatically clean data, select relevant features, generate analysis-ready scripts across multiple platforms, and create interactive dashboards — reducing manual work and making analytics faster and more accessible.

Scope

1. Automated Hybrid Cleaning Develop a high-performance system using Polars/Pandas for speed and Random Forest/KNN for context-aware imputation. It automates data sanitation by dynamically switching between rule-based and ML-driven strategies to fix missing values and detect anomalies (Isolation Forest).
2. Semantic Feature Engineering Build an AI-enhanced engine utilizing Sentence Transformers and Google Gemini to synthesize relevant measures and filter features. It combines statistical rigor with business reasoning to ensure dashboard readiness and maximize analytical depth.
3. Automatically generates and executes analysis-ready scripts in Python, R, and SQL using LLM, along with Excel-based analytical logic using Pandas, enabling users to obtain insights through natural-language queries without extensive coding effort.
4. Creates editable and downloadable Excel dashboards using XlsxWriter, Openpyxl, and Pywin32, incorporating pivot tables and charts to support flexible customization for business reporting and decision-making.

Technological Stack

1. Data Preprocessing and Cleaning

- **Libraries/Tools:**
- **Pandas & Polars** → Used for structured data manipulation such as merging, grouping, and reshaping, while **Polars** enables rapid ingestion and low-latency processing of large-scale datasets with superior performance.
- **KNN Imputer** → Preserves local data correlations in numerical gaps.
- **Random Forest** → Predicts missing categorical values.
- **Role:** Achieves low-latency performance by utilizing Polars for rapid ingestion of large-scale structured datasets.
- **2. Feature Selection and Extraction**
- **Libraries/Tools**: SentenceTransformers, Google Gemini SDK.
  - **Scikit-learn** → Enables intelligent data imputation and anomaly detection through efficient machine-learning algorithms for reliable preprocessing.
  - **Gemini 2.5 Flash** → LLM reasoning to validate feature utility and synthesize business measures.
  - **all-MiniLM-L6-v2** → Vector model for semantic schema mapping

Technological Stack

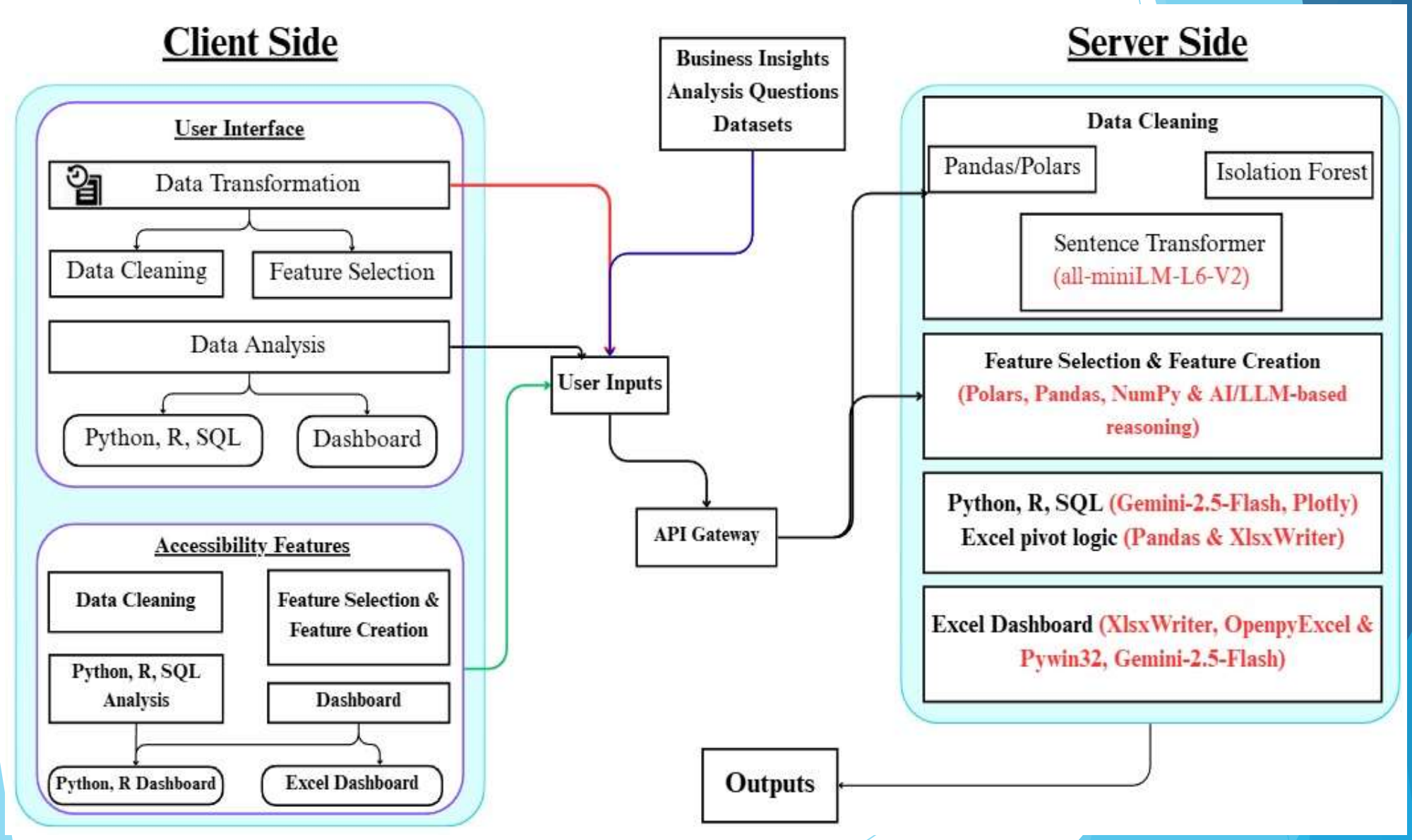
3. Automated Script Generation

- **Technologies:**
  - **Python** → Core engine for automation.
  - **R** → Optional statistical scripting support.
  - **SQLAlchemy** → For SQL query generation and execution.
- **Methodology:** Convert user requirements → automatically generate code (Python, R, SQL, or hybrid).

4. Dashboard Creation & Visualization

- **Libraries/Tools:**
  - **XlsxWriter** → For editable Excel dashboards.
  - **Matplotlib / Seaborn / Plotly** → For charts & plots.
  - **Pivot Table (Excel)** → For interactive analysis.
- **Methodology:** Auto-generate downloadable dashboards with layout suggestions.

Proposed system architecture/Working/Module Diagram



Prototype Design Demonstration

Implementation Status

Review Suggestions (Given in Last meeting)

- **Take 2nd paper as base paper and place it 1st**
- **Need to review the literature survey**
- **Preprocessing steps needs to be studied**
- **Implementation should be started**
- **Modify the system diagram**

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Thank You...!!